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2027 CYCLE

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CUET UG

Chemistry

d and f Block Elements

Official Syllabus Unit IV · Class 12 · CUET UG 2027 Cycle Notes

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) Chemistry syllabus 2026, the current 2027 baseline, Unit IV)	Skills the questions test
Position and configuration	General introduction to d and f-block elements, Position of d- and f-block elements in the periodic Table, Electronic configurations of the d- and f-block elements	Writing $(n-1)d^{1-10}ns^{1-2}$ configurations, spotting the Cr and Cu exceptions, and deciding why Zn, Cd, Hg sit outside the strict "transition element" club.
General trends of transition elements	General properties of the transition elements & their trends: Physical properties, atomic & ionic sizes - lanthanoid contraction, Ionisation Enthalpies, Oxidation States, Standard electrode potential trends for M2+/M & M3+/M2+, Stability of Higher oxidation states, Chemical reactivity and E° values	Explaining size and ionisation trends, listing the oxidation-state ladder of the 3d series, and reading E° values (including the copper anomaly) correctly.
Magnetism, colour, complexes, catalysis	Magnetic properties, Formation of coloured Ions, Formation of Complex compounds, Catalytic properties, Formation of Interstitial Compounds, Alloy Formation	Computing spin-only moments from unpaired electrons, deciding which ions are coloured, and naming the standard catalyst, interstitial and alloy examples.
Oxides and key compounds	Nature of Oxides & Oxanions of d-block Metals, Some important compounds of transition elements- K2Cr2O7 and KMnO4	Ordering oxide acidity across the series and with oxidation state, and reproducing the chromate/dichromate switch and the permanganate half-equations in acid and alkali.
The f-block	Lanthanoids & actinoids: general properties, Atomic & Ionic sizes, Oxidation States, Comparison of actinoids with Lanthanoids, Applications of d-and f-Block elements	Stating the lanthanoid contraction and its consequences, contrasting actinoid and lanthanoid behaviour, and

Sub-topic	Official syllabus pointer (NTA CUET(UG) Chemistry syllabus 2026, the current 2027 baseline, Unit IV)	Skills the questions test
		naming everyday applications.

EXAM PATTERN

CUET UG pattern (NTA official information bulletin and syllabus, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every correct answer and −1 for every incorrect answer. The Chemistry paper maps strictly to NCERT Class 12, and Unit IV is the most fact-dense unit on the paper: configurations, trends, colours and the two named workhorse compounds.

The d-Block: Where the Periodic Table Gets Interesting

Sweep your finger across the middle of the periodic table and you cross the industrial heart of chemistry — iron rails, copper wires, titanium jet engines, the platinum in a car's catalytic converter. These are the **d-block elements**: their distinguishing electron lands in the second-outermost shell, the $(n-1)d$ subshell, while the outermost ns fills first. That odd "fill the inside shell last" habit gives them a partly filled d subshell in most states — and every famous property of this chapter flows from that single fact: variable oxidation states, colour, magnetism, catalysis.

- **Position:** groups 3 to 12, sandwiched between the s-block (left) and p-block (right). The d-block plus group 12 are collectively the **transition metals** in casual talk — but the strict definition is tighter (below).
- **Three series:** 3d (Sc to Zn), 4d (Y to Cd), 5d (La, then Hf to Hg — with the f-block ducking out in between). A 6d series begins at Ac.
- **General configuration:**

$$(n-1)d^{1-10}ns^{1-2}$$

  — note the d count starts from 1 (Sc: $3d^1 4s^2$), never zero.
- **The two famous exceptions:** chromium takes $3d^5 4s^1$ and copper $3d^{10}4s^1$ — each steals its 4s electron because a **half-filled** or **fully-filled** d subshell carries extra stability. These two configurations are guaranteed one-mark questions.
- **The strict definition:** a **transition element** is one whose atom **or a common ion** has a partly filled d subshell. Zinc, cadmium and mercury end with d^{10} in the atom **and** in their only common ions (Zn^{2+} is $3d^{10}$), so they are d-block but not truly transition — and they predictably lack colour, variable states and catalytic superpowers.

Element	Configuration written	Why / consequence
Sc (Z = 21)	$3d^1 4s^2$	First of the 3d series; only the +3 state in practice ($3d^0$, colourless)

Element	Configuration written	Why / consequence
Cr (Z = 24)	$3d^5 4s^1$	Half-filled d subshell — extra stability; 6 unpaired electrons in the atom
Cu (Z = 29)	$3d^{10} 4s^1$	Fully-filled d subshell; explains Cu(II) being $3d^9$ and coloured while Cu(I) is $3d^{10}$
Zn (Z = 30)	$3d^{10} 4s^2$	Full d in atom and in Zn^{2+} — no d–d transitions, no colour, no variable states

Physical and Size Trends Across the Series

- **All metals, properly hard:** dense, lustrous, high melting points, good conductors — the d electrons join the metallic bonding party, gluing atoms tightly together.
- **Melting points climb, peak, then dip:** strength of metallic bonding grows as unpaired d electrons increase, peaks near the middle (tungsten holds the all-time record among metals; in the 3d row, chromium and manganese region peak around vanadium), then falls as d-electron pairing sets in. Manganese is the classic anomaly — its $3d^5 4s^2$ atoms bond weakly (stable configuration, unwilling to share), so Mn melts far lower than its neighbours.
- **Atomic size shrinks, then swells:** across the 3d row, added d electrons screen each other poorly, so size contracts up to about chromium; near the end, electron–electron repulsion inside the crowded d subshell wins and the size ticks back up (Zn > Cu).
- **Ionic size tracks charge:** for the same element, higher oxidation state means smaller ion — Fe^{3+} (0.65 Å) < Fe^{2+} (0.76 Å). More positive charge pulls the electron cloud in.
- **Density rises across each series:** shrinking size plus growing mass — osmium and iridium are the density champions of the entire periodic table.

The lanthanoid contraction: a shrinkage with consequences

- **What it is:** the steady, cumulative decrease in atomic and ionic radii from lanthanum to lutetium, caused by the poor shielding of nuclear charge by 4f electrons — each added 4f electron pulls the outer shell in slightly, and the effect adds up over fourteen elements.
- **Consequence 1:** the 5d series (after the f-block) starts at nearly the **same size** as the 4d series above it. Zirconium and hafnium become near-perfect size twins (Zr 160 pm, Hf 159 pm) — which is why their chemistry is so similar that separating them was a chemical nightmare.

- **Consequence 2:** the 5d elements are denser and more electropositive than their 4d twins, and the +3 ions of the lanthanoids themselves differ so little in size that separating individual lanthanoids is famously hard.
- **Consequence 3:** basic strength of lanthanoid hydroxides **decreases** from

$$\mathrm{La(OH)_3}$$

to

$$\mathrm{Lu(OH)_3}$$

— smaller ions hold their hydroxides more covalently.

Ionisation Enthalpies and Oxidation States

- **Ionisation enthalpy climbs across a series** — the shrinking radius holds electrons tighter — with small dips (Cr, Cu) where the d subshell is half/full and the electron hands over easily.
- **Second and third ionisations reveal the *ns* electrons leave first:** removing 4*s* costs little; once the *d* subshell is being emptied, the costs jump. That staircase is the mechanical reason variable oxidation states exist at all.
- **Why so many oxidation states?** The 4*s* and 3*d* electrons sit at nearly the same energy, so the element can lose or share any number from 0 to 7 — the whole ladder sits within reach. The p-block cannot do this (its p electrons only), and neither can the s-block.

Element	States seen	Element	States seen
Sc	+3 only	Fe	+2, +3 (also +6 in ferrates)
Ti	+2, +3, +4	Co	+2, +3
V	+2, +3, +4, +5	Ni	+2, +3 (rare)
Cr	+2, +3, +6	Cu	+1, +2
Mn	+2, +3, +4, +6, +7	Zn	+2 only (d^{10})

- **Manganese is the spread champion** (+2 to +7) because its d^5 heart can shed or share electrons in neat steps; Sc and Zn bookend the row with a single state each ($3d^0$ and $3d^{10}$ — nothing left to vary).
- **Middle states rule the 3d row:** +2 and +3 dominate. The

$$\mathrm{M^{2+}}$$

state is easiest to reach (4s electrons go first) and stays common for every element from Ti to Cu.
- **Down a group, higher states gain stability** — the exact **opposite** of the p-block trend. Chromium(VI) is a fierce oxidiser (it wants to fall to +3), while molybdenum(VI) and tungsten(VI) are stable and peaceful. Same story down group 7: Mn(VII) oxidises, Tc(VII) and Re(VII) barely bother.

Standard Electrode Potentials and Reactivity

- **The**

  $$\mathbf{M^{2+}/M}$$

  ladder: most 3d metals sit at negative potentials (Zn −0.76 V, Ni −0.25 V) — they dissolve in acids. **Copper is the exception** at +0.34 V: its full cost sheet (atomisation + two ionisations, with poor hydration payback) exceeds the refund, so copper refuses dilute acids like HCl and

  $$H_2SO_4$$

  (oxidising acids such as

  $$HNO_3$$

  still attack it, by a different route).
- **A general zigzag:** the values do not fall smoothly — V, then a rise, dips at Cu, rises again to Zn — because each value bundles three different costs (atomisation, ionisation, hydration) that trend differently.
- **The**

  $$\mathbf{M^{3+}/M^{2+}}$$

  ladder: here the picture is clean and useful:
  - **Sc^{3+} is the most stable +3:**

    $$Sc^{3+}$$

    ($3d^0$) never reverts — the

    $$Sc^{3+}/Sc^{2+}$$

    couple has no meaningful existence.
  - **Fe^{3+}/Fe^{2+} at +0.77 V:** moderately strong oxidiser —

    $$Fe^{3+}$$

    loves returning to the extra-stable half-filled $3d^5$ of

    $$Fe^{2+}$$

    … in fact

    $$Fe^{2+}$$

    ($3d^6$) is **less** stable, so the couple reads positive.
  - **Mn^{3+} (+1.57 V) and Co^{3+} (+1.97 V) are the fierce ones:** they are desperate to reach $3d^5$ (Mn^{2+}) and $3d^6$-octahedral-low-spin (Co^{3+}) respectively — hence Mn(III) and Co(III) compounds oxidise water itself.
  - **V^{3+}/V^{2+} at −0.26 V:** vanadium's +2 is a happy $3d^3$… and the couple stays negative —

    $$V^{2+}$$

    is a gentle reducing agent, no rush to oxidise.

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- **Chemical reactivity, row by row:** the 3d metals are **more** reactive than 4d/5d cousins (smaller size, better access). Chromium and nickel passivate — a thin oxide film shuts down further attack (that is why chromium plating protects bumpers).

Magnetism and Colour: The d-Electron Light Show

Magnetic properties

- **Unpaired d electrons = paramagnetism.** Each unpaired electron carries a magnetic moment; paired ones cancel. Compute the **spin-only** moment:

$$\mu = \sqrt{n(n+2)}\ \text{BM}$$

with n = number of unpaired electrons. Iron(III), $3d^5$, has $n = 5$:

$$\mu = \sqrt{35} \approx 5.92\ \text{BM}$$

— the maximum in the row.

- **Measure and you can count:** a magnetic measurement is a hidden electron-counting exercise — it distinguishes

$$\text{Fe}^{3+}(d^5)$$

from

$$\text{Fe}^{2+}(d^6)$$

, or tells you a complex is low-spin.

Ion	d-count, unpaired, moment	Ion	d-count, unpaired, moment
Ti^{3+}	d^1, 1, 1.73 BM	Mn^{2+}	d^5, 5, 5.92 BM
V^{3+}	d^2, 2, 2.83 BM	Fe^{2+}	d^6, 4, 4.90 BM
Cr^{3+}	d^3, 3, 3.87 BM	Co^{2+}	d^7, 3, 3.87 BM
Mn^{3+}	d^4, 4, 4.90 BM	Ni^{2+}	d^8, 2, 2.83 BM
Cr^{2+}	d^4, 4, 4.90 BM	Cu^{2+}	d^9, 1, 1.73 BM
Mn^{4+}…	d^3 (in MnO_2), 3	Zn^{2+}	d^{10}, 0, 0 BM (diamagnetic)

Colour from d–d transitions

- **The mechanism:** in a complex, the five d orbitals split into two energy sets. White light kicks a d electron from the lower set to the upper one — a **d–d transition** — and the light that is **not** absorbed gives the ion its colour. Blue aqueous

$$\text{Cu}^{2+}$$

absorbs red-orange light; the complementary blue reaches your eye.

- **No partly filled d, no colour:**

$$Sc^{3+}(d^0)$$

,

$$Ti^{4+}(d^0)$$

,

$$Zn^{2+}(d^{10})$$

,

$$Cu^{+}(d^{10})$$

— all colourless. There is no electron to promote, or no empty d slot to promote it into.

- **The charge-transfer surprise:**

$$MnO_4^-$$

(deep purple) and

$$CrO_4^{2-}$$

(bright yellow) contain Mn(VII) and Cr(VI) — both $3d^0$: **no d electrons at all**, yet they are among the most colourful ions in chemistry. Their colour comes from **charge transfer** — an electron momentarily hopping from oxygen to the metal. Remember this exception; it is a beloved MCQ.

Complexes, Catalysts, Interstitials and Alloys

- **Complex formation:** transition ions are small, highly charged, and own vacant d orbitals that can accept lone pairs from ligands — so coordination numbers of 4 and 6 come naturally: $[Fe(CN)_6]^{4-}$, $\left[Cu(NH_3)_4\right]^{2+}$, $[Ni(CO)_4]$. The main-group ions (like

$$Na^+$$

) rarely bother.
- **Catalytic behaviour:** the same flexibility powers catalysis — surface atoms accept and hand back reactant molecules, or cycle through oxidation states. The syllabus's standard set: **Fe** (Haber process,

$$NH_3$$

), $\mathbf{V_2O_5}$ (contact process,

$$H_2SO_4$$

), **Ni** (vegetable-oil hydrogenation), $\mathbf{TiCl_4}$ **with** $\mathbf{AlMe_3}$ (Ziegler–Natta polythene), **Pd/ Pt/ Rh** (car exhaust converters).
- **Interstitial compounds:** small atoms — H, C, N — slip into the holes of the metal lattice without changing its shape. Products stay metallic: hard, dense, electrically conducting, and **non-stoichiometric** (steel is Fe with interstitial C at a few per cent; cast iron has more). These compounds are chemically inert yet very hard — steel nails, drill bits.
- **Alloy formation:** transition metals have similar radii, so one metal's atoms substitute freely in another's lattice — brass (Cu + Zn), bronze (Cu + Sn), stainless steel (Fe + Cr +

Ni), ferrochrome. The 4d/5d twins (Zr/Hf) alloy with each other almost indistinguishably — a direct echo of the lanthanoid contraction.

Oxides and Oxoanions: Acidity Follows Charge

- **Across the series (same oxidation state):** oxides turn from basic to amphoteric to acidic — MnO is basic, "FeO" basic-ish, CrO amphoteric, Cr_2O_3 amphoteric — the clean exam-ready ladder is for **manganese**: MnO (basic) → Mn_3O_4 → Mn_2O_3 → MnO_2 (amphoteric) → Mn_2O_7 (strongly acidic, gives $HMnO_4$).
- **The rule that matters: higher oxidation state = more acidic oxide.** The +7 metal in Mn_2O_7 pulls electron density so hard that its O–H bonds turn fully acidic. Low states keep ionic, basic character.
- **Oxoanions carry the top states:** VO_2^+ (V), CrO_4^{2-}, $Cr_2O_7^{2-}$ (Cr), MnO_4^- (Mn) — stable, colourful, and the homes of the highest oxidation states in aqueous chemistry.

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The Two Workhorses: $K_2Cr_2O_7$ and $KMnO_4$

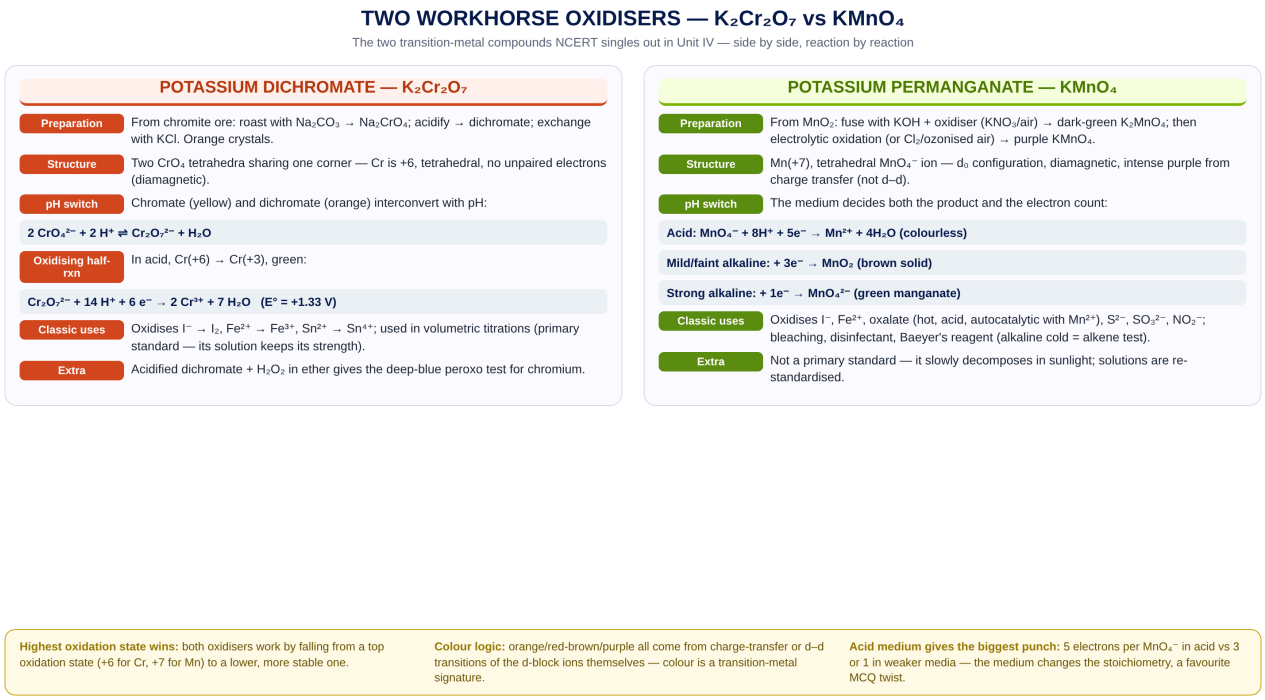


Figure 1: The two named compounds side by side: preparations, structures, pH-switched equilibria, oxidising half-equations and classic uses of potassium dichromate and potassium permanganate.

- **Potassium dichromate, preparation line:** chromite ore

$$(\mathrm{Fe\ Cr_2O_4})$$

roasted with sodium carbonate → sodium chromate; acidified → dichromate; crystallised with KCl. The chromate–dichromate switch is a pure pH lever:

$$2\mathrm{CrO_4^{2-}} + 2H^+ \rightarrow \mathrm{Cr_2O_7^{2-}} + \mathrm{H_2\ O}$$

Alkali pushes it left (yellow chromate); acid pushes it right (orange dichromate). Same chromium, same charge — just pH.

- **Dichromate as oxidiser (acid medium):** Cr(VI) → Cr(III), orange to green, three electrons per chromium:

$$\mathrm{Cr_2O_7^{2-}} + 14H^+ + 6e^- \rightarrow 2\mathrm{Cr^{3+}} + 7\mathrm{H_2\ O}$$

It oxidises I^- to I_2,

$$\mathrm{Fe^{2+}}$$

to

$$\mathrm{Fe^{3+}}$$

,

$$\mathrm{Sn^{2+}}$$

to

$$\mathrm{Sn^{4+}}$$

— and because its solution is stable, it serves as a **primary standard** in titrations.

- **Potassium permanganate, preparation line:** MnO_2 fused with KOH and an oxidant (air/KNO_3) → green manganate

$$MnO_4^{2-}$$

; electrolytic oxidation → purple

$$MnO_4^-$$

.

- **Permanganate's pH-dependent strength** — the same ion delivers a different electron count in each medium:

$$\text{Acid}: MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O \quad (5 \text{ electrons})$$

$$\text{Mild alkaline}: MnO_4^- + 2H_2O + 3e^- \rightarrow MnO_2 + 4OH^- \quad (3 \text{ electrons})$$

$$\text{Strong alkaline}: MnO_4^- + e^- \rightarrow MnO_4^{2-} \quad (1 \text{ electron})$$

- **Permanganate's classic jobs:** titrating Fe^{2+} and oxalate (hot, acidified, slow until Mn^{2+} appears — **autocatalysis**), the Baeyer alkene test (cold alkaline

$$KMnO_4$$

decolourised by C=C), disinfecting water. Unlike dichromate it is **not** a primary standard — sunlight slowly decomposes it.

The f-Block: Lanthanoids and Actinoids

Pull the fourteen elements from cerium to lutetium (plus lanthanum) out of period 6 and park them in a row below the table — that is the **lanthanoid** series, filling the inner 4f subshell. The **actinoids** (thorium to lawrencium, filling 5f) mirror them in period 7. They are "inner transition elements" — the transition metals' transition metals.

Lanthanoids: the +3 family

- **Configurations:** $[Xe]4f^{1-14}5d^{0-1}6s^2$ — the 4f electrons sit deep inside, shielded by the 5s and 5p shells, which is why f-electron chemistry is so uniform.
- **Oxidation states:** +3 dominates across all of them. The two honoured exceptions follow the half-filled/full-filled logic you already met in the d-block:

$$Eu^{2+}$$

($4f^7$, half-filled — stable) and

$$Tb^{4+}$$

($4f^7$); plus

$$Ce^{4+}$$

($4f^0$, empty — stable) and

$$Yb^{2+}$$

($4f^{14}$).

- **Physical look:** silvery-white soft metals, high melting points, very dense; slow tarnishing in air.
- **Sizes shrink steadily** — the **lanthanoid contraction** explained in the d-block section — and ionic radii change so little between neighbours that separating adjacent lanthanoids took chemists decades of ion-exchange grinding.
- **Colours and magnetism return:** f–f transitions give many lanthanoid ions lovely colours, and unpaired f electrons give paramagnetism — but the f electrons are so deeply buried that bonding is almost entirely ionic.
- **Applications:** mischmetal (a cerium-rich alloy, 95% lanthanoids + iron) sparks lighter flints; neodymium magnets are the strongest permanent magnets sold; cerium oxide polishes glass; thorium feeds experimental nuclear reactors.

Actinoids: the radioactive wild cousins

- **Configurations:** $[\mathrm{Rn}]5f^{1-14}6d^{0-1}7s^2$ — the 5f subshell is bigger, more diffuse, and **participates in bonding**, unlike the tucked-away 4f.
- **Oxidation states sprawl from +3 to +7** (neptunium and plutonium reach +7) — the direct consequence of 5f's looseness. Every actinoid is radioactive; none beyond uranium occurs naturally in quantity.
- **The comparison the syllabus wants:**

Property	Lanthanoids	Actinoids
Filling	4f, well shielded	5f, less shielded, more involved
Oxidation states	+3 overwhelmingly (+2, +4 rarities)	+3 to +7, often several at once
Radioactivity	Only promethium (and late members)	All of them, all the time
Chemical variety	Near-identical ionic chemistry	Rich covalent, complex-forming chemistry
Bonding	Almost purely ionic	Significant covalency in high states

Key Reactions & Rules: One-Stop Revision

CONFIGURATIONS, MOMENTS, EQUATIONS

Rule / formula	Symbols in plain words	When to reach for it
$(n-1)d^{1-10}ns^{1-2}$	d-block filling pattern; the ns pair fills first	Every configuration question

Rule / formula	Symbols in plain words	When to reach for it
Cr: $3d^5 4s^1$; Cu: $3d^{10} 4s^1$	half/full-filled d stability	The guaranteed one-mark exceptions
$\mu = \sqrt{n(n+2)}$ BM	n = unpaired electrons; Fe^{3+} gives 5.92	Spin-only moment calculations
Coloured → partly filled d	d–d transitions absorb light	Colour predictions (Sc^{3+}, Zn^{2+} colourless)
Charge-transfer colour	MnO_4^- purple, CrO_4^{2-} yellow with d^0 ions	The colour-without-d-electrons trap
$Fe^{3+} < Fe^{2+}$ size	higher charge, smaller ion	Ionic radius comparisons
Higher OS → more acidic oxide	MnO basic → Mn_2O_7 acidic	Oxide acidity ladders
Down a group: high OS stabilises	opposite of p-block	CrO_4^{2-} vs MoO_4^{2-}/WO_4^{2-} checks
$2CrO_4^{2-} + 2H^+ \rightarrow Cr_2O_7^{2-} + H_2O$	yellow → orange, pH lever	Chromate–dichromate conversions
$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$	orange → green, $E^\circ = +1.33$ V	Dichromate titrations
$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$	acid: 5 electrons, purple → colourless	Permanganate in acid medium
$MnO_4^- + 3e^- \rightarrow MnO_2$ (mild alkali); $+e^- \rightarrow MnO_4^{2-}$ (strong alkali)	3 and 1 electrons	Permanganate in weak/strong alkali
Lanthanoid contraction	poor 4f shielding; Zr ≈ Hf sizes	Size and consequence questions
Eu^{2+} $(4f^7)$, Tb^{4+} $(4f^7)$, Ce^{4+} $(4f^0)$	half/full/empty f-shell logic	Lanthanoid oxidation-state exceptions

Exceptions & Traps

THE TRAPS THIS CHAPTER SETS — READ BEFORE THE MOCK

The rule you assume	The exception / twist	The trap in the question
All d-block elements are "transition"	Zn, Cd, Hg have d^{10} atoms and d^{10} ions	The definition demands a **partly filled** d subshell in atom or common ion
4s fills before 3d, so 4s fills first in ions too	Ions lose the 4s electrons FIRST	Fe^{2+} is $3d^6$ (not $3d^4\,4s^2$); Fe^{3+} is $3d^5$
Down the group, lower states stabilise (p-block habit)	d-block is the opposite: higher states stabilise down	Mn(VII) oxidises; Tc(VII), Re(VII) do not
Colour always needs d electrons	MnO_4^- and CrO_4^{2-} are d^0 yet intensely coloured	Charge-transfer absorption — the classic "how is d^0 purple?" question
Cu dissolves in dilute acids like other metals	$E°(Cu^{2+}/Cu) = +0.34$ V blocks H^+	Copper needs oxidising acids (HNO_3) — dilute HCl/H_2SO_4 do nothing
Lanthanoids show many oxidation states like actinoids	Lanthanoids are +3 monogamists	Actinoids sprawl to +7; the comparison table decides such questions
All purple/coloured ions are d–d coloured	Charge transfer paints MnO_4^-; the medium changes its electron count	5 electrons in acid, 3 in mild alkali, 1 in strong alkali
Interstitial compounds are new stoichiometric compounds	They are non-stoichiometric	Steel's carbon content varies continuously — no fixed formula

The Whole Chapter in One Picture

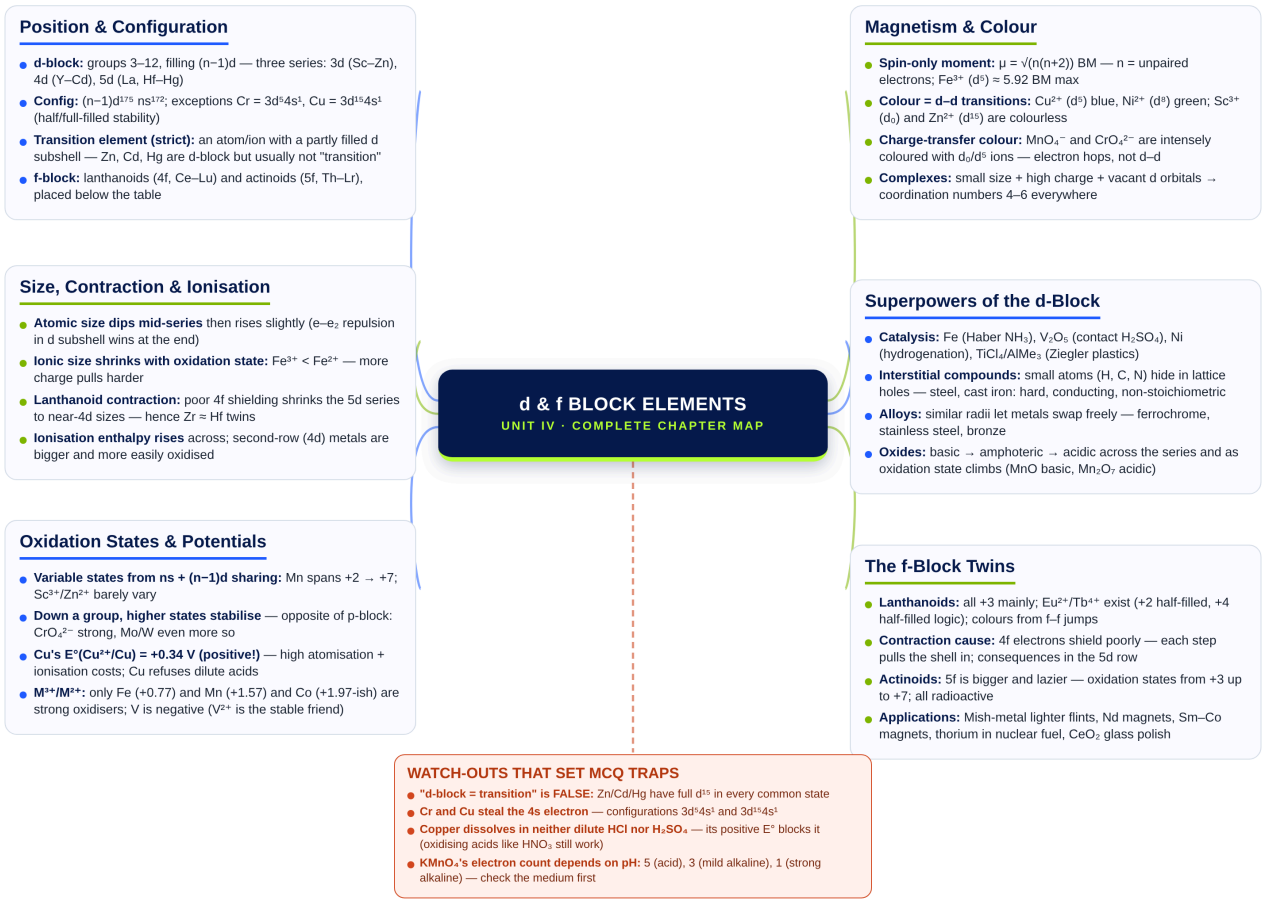


Figure 2: Chapter mindmap: position and configurations, sizes and the lanthanoid contraction, oxidation states and electrode potentials, magnetism and colour, catalysis-interstitials-alloys, and the f-block twins — with the traps that catch most students.

Practice Questions (Exam Pattern)

Ten questions in the CUET UG style. Attempt all of them — the paper gives +5 for each correct answer and −1 for every incorrect answer.

Q1. Which of the following electronic configurations corresponds to chromium (Z = 24)?

(A) $[\mathrm{Ar}]3d^4 4s^2$ **(B)** $[\mathrm{Ar}]3d^5 4s^1$
(C) $[\mathrm{Ar}]3d^6 4s^0$ **(D)** $[\mathrm{Ar}]3d^3 4s^2 4p^1$

Q2. Zn, Cd and Hg are generally NOT regarded as transition elements because:

(A) they are soft metals

(B) their atoms and common ions have completely filled d^{10} subshells

(C) they show only +2 oxidation states

(D) they do not conduct electricity

Q3. The magnetic moment (spin-only) of Fe^{3+} ($3d^5$) is about:

(A) 1.73 BM **(B)** 3.87 BM
(C) 4.90 BM **(D)** 5.92 BM

Q4. Which aqueous ion is colourless?

(A) Cu^{2+} **(B)** Ni^{2+}
(C) Zn^{2+} **(D)** Fe^{3+}

Q5. The deep purple colour of MnO_4^- (Mn in +7, $3d^0$) is due to:

(A) d–d transitions **(B)** charge transfer between oxygen and manganese
(C) unpaired electrons **(D)** metal–metal bonds

Q6. In acidic medium, one MnO_4^- ion accepts how many electrons?

(A) 1 **(B)** 3
(C) 5 **(D)** 7

Q7. The lanthanoid contraction is responsible for:

(A) the +3 state of lanthanoids **(B)** Zr and Hf having almost identical radii
(C) the colour of Cu^{2+} **(D)** the acidity of Mn_2O_7

Q8. When the pH of a chromate/dichromate solution is lowered (acid added), the equilibrium:

(A) shifts to yellow chromate **(B)** shifts to orange dichromate
(C) does not shift **(D)** precipitates chromium metal

Q9. Which pair shows the p-block-contrary trend "higher oxidation states become more stable DOWN the group"?

(A) Pb^{2+} vs Pb^{4+} **(B)** Cr(VI) vs Mo(VI) vs W(VI)
(C) Na^+ vs K^+ **(D)** Cl^- vs Br^-

Q10. Steel and cast iron are everyday examples of:

(A) alloys formed by substitution **(B)** interstitial compounds
(C) coordination compounds **(D)** charge-transfer complexes

Detailed Solutions

Sol. Q1. Chromium steals one 4s electron into 3d: $[Ar]3d^5 4s^1$ — the half-filled d subshell plus single 4s electron sits lower in energy than $3d^4 4s^2$. Copper does the same for a **fully** filled $3d^{10}$.

Answer: (B)

Sol. Q2. The strict definition needs a **partly filled** d subshell in the atom or a common ion. Zn^{2+} is $3d^{10}$ — full — so no d–d transitions, no colour, no variable states: d-block, yes; transition, no.

Answer: (B)

Sol. Q3. Spin-only formula with $n = 5$: $\mu = \sqrt{5 \times 7} = \sqrt{35} \approx 5.92$ BM. Memorise the ladder: $1 \to 1.73$, $2 \to 2.83$, $3 \to 3.87$, $4 \to 4.90$, $5 \to 5.92$.

Answer: (D)

Sol. Q4. Zn^{2+} is $3d^{10}$ — no electron can jump between d levels, so no d–d absorption, so colourless. Cu^{2+} (blue), Ni^{2+} (green) and Fe^{3+} (yellow-brown in water) all have partly filled d subshells.

Answer: (C)

Sol. Q5. Mn(VII) has zero d electrons — no d–d transition is possible. The purple comes from charge transfer: an oxygen electron momentarily hops to manganese. Same story for yellow CrO_4^{2-}.

Answer: (B)

Sol. Q6. Acid medium:

$$MnO_4^- + 8H^+ + 5e^- \to Mn^{2+} + 4H_2\,O$$

— five electrons, the maximum punch. Mild alkali gives 3 (to brown MnO_2); strong alkali gives 1 (to green manganate).

Answer: (C)

Sol. Q7. Fourteen poorly-shielding 4f electrons shrink the lanthanoids cumulatively, so the 5d row starts at 4d-sized radii — Zr (160 pm) and Hf (159 pm) become twins. The +3 state, Cu^{2+}'s colour and Mn_2O_7's acidity live elsewhere in the chapter.

Answer: (B)

Sol. Q8. Lower pH means more H^+, so the equilibrium

$$2CrO_4^{2-} + 2H^+ \to Cr_2O_7^{2-} + H_2\,O$$

shifts right — yellow chromate turns orange dichromate. Adding alkali reverses it.

Answer: (B)

Sol. Q9. The d-block defies the p-block habit: down group 6, Cr(VI) is a strong oxidiser but Mo(VI) and W(VI) are stable and mild — higher states stabilise down the group. Pb^{2+}/Pb^{4+} shows the **p-block** trend (inert pair).

Answer: (B)

Sol. Q10. Carbon atoms sit in the **holes** of the iron lattice — interstitial compounds: hard, conducting, non-stoichiometric. Brass and bronze are substitutional alloys; coordination complexes are a different family entirely. The same interstitial idea explains why steel drills cut where pure iron bends.

Answer: (B)

Wrong-Answer Clinic — the one-line corrections

Dropped marks above? Find your slip here and rewrite the corrected rule in your own words.

If you answered...	Your slip was...	The corrected rule, in one line
Q1 option A	Filling 3d before stealing 4s	Half-filled beats regular: Cr keeps only ONE 4s electron — $3d^5 4s^1$
Q3 option C	Using n = 4 in the formula	Fe^{3+} is $3d^5$ — five unpaired electrons, $\sqrt{35} \approx 5.92$ BM, the row's maximum
Q6 option B	Mixing media	Count electrons by medium: acid 5, mild alkali 3, strong alkali 1 — always read the medium first
Q7 option A	Confusing the cause and the effect	The contraction is a **size** story (poor 4f shielding); its twin-effect headline is Zr ≈ Hf
Q9 option A	Applying the p-block habit	Lead follows the inert-pair rule (p-block); only d-block rows stabilise high states downward
Q8 option A	Reading "lower pH" as "more alkali"	More acid pushes $2CrO_4^{2-} + 2H^+ \rightarrow Cr_2O_7^{2-} + H_2O$ to the RIGHT: yellow orange
Q10 option A	Calling steel an alloy of substitution	Steel's carbon hides in lattice holes (interstitial); brass/bronze are the substitutional alloy examples

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